Modeling and Forecasting Monthly Atmospheric CO_2 at Mauna Loa, 1958–2026

Final Project Report

Pascal Antwi · STA6938 Time Series · Instructor: Y. Jia · December 6, 2026

Abstract. This report analyzes 822 monthly mean concentrations of atmospheric carbon dioxide at Mauna Loa, March 1958 to August 2026. The series has an accelerating trend and a stable 12-month cycle, so regular and seasonal differences are required. Of four model families compared (ARMA, ARIMA, seasonal ARIMA, and regression with autocorrelated errors), the ACF/PACF, information criteria and residual checks select the seasonal ARIMA$(0,1,1)\times(0,1,1)_{12}$. Both parameters are highly significant, the model is invertible, and its residuals are white noise with constant variance. Across 276 rolling out-of-sample forecasts it beats the other model families, and it forecasts that monthly CO_2 will first exceed 440 ppm in April 2030.

Keywords: seasonal ARIMA; ARMA; regression with autocorrelated errors; Holt–Winters; Mauna Loa.

1. Background and Motivation

Carbon dioxide is the most important long-lived greenhouse gas, and its rising concentration is the main driver of global warming. The Mauna Loa record, started by Charles D. Keeling in 1958, is the longest continuous direct measurement of it [1], and climate targets are framed around how fast CO_2 rises and when it crosses particular thresholds. I chose this topic because climate change is one of the defining problems of my generation, and I wanted my project to answer a question with real consequences rather than analyze a textbook example. The series is also statistically rich, combining a trend, a seasonal cycle and short-term dependence.

That CO_2 is rising is well known; how fast it is rising, whether the rise is accelerating, and what levels to expect in the next few years must be quantified with proper statistical uncertainty. The objectives are to (i) characterize the trend and seasonality of monthly CO_2; (ii) build and validate an adequate time series model, comparing ARMA, ARIMA, seasonal ARIMA and regression with autocorrelated errors; (iii) measure whether growth is accelerating; and (iv) forecast monthly CO_2 to 2030 with prediction intervals, including when it will first exceed 440 ppm.

2. Description of the Series

The data are the monthly mean CO_2 mole fractions, in parts per million (ppm), published by the NOAA Global Monitoring Laboratory [2] and freely available at https://gml.noaa.gov/ccgg/trends/data.html. The series X_t contains $n = 822$ consecutive months from March 1958 to August 2026, with no missing values (Figure 1 and Table 1). Concentration rose from 315.71 ppm in March 1958 to 427.55 ppm in August 2026, an increase of 111.8 ppm (35%). The average 12-month increase over the whole record is 1.68 ppm per year, but this rate has more than tripled, from 0.86 ppm/yr in the 1960s to 2.40 ppm/yr in the 2010s and 2.54 ppm/yr in the 2020s (Figure 2 and Table 2). The largest 12-month increases occurred in March 2024 (4.39 ppm) and April 2016 (4.25 ppm), both El Niño years.

A classical additive decomposition with a 2×12 moving-average filter [3] (Figure 3) attributes 96.6% of the variance of X_t to the trend. The seasonal component peaks in May (+3.04 ppm) and reaches its minimum in October (-3.25 ppm), an amplitude of 6.29 ppm (Figure 4 and Table 3). This reflects uptake by Northern Hemisphere vegetation in the growing season; the irregular component is small (SD 0.30 ppm).

3. Stationarity of the Series

Time plot. The first step was the time plot (Figure 1), which shows two clear features: a persistent upward trend, and a regular 12-month seasonal cycle with nearly the same shape every year. The mean of X_t therefore depends on t, and the series is non-stationary.

Transformation. A log or Box–Cox transformation is warranted when variability grows with the level [3]. The within-year range of the detrended series (Figure 5) stays between 5.3 and 7.6 ppm (6.0 ppm in the 1960s, 7.0 ppm in the 2010s) while the level rose 35%, so the additive form is adequate and no transformation is used.

Sample ACF. The sample ACF agrees (Figure 6): $\hat{\rho}(1) = 0.996$, and the ACF is still 0.865 at lag 36. After a first difference, ∇X_t, the ACF at the seasonal lags 12, 24 and 36 remains 0.925, 0.910 and 0.894. This failure to decay shows that the seasonal pattern is also non-stationary, so a seasonal difference $\nabla_{12} = 1 - B^{12}$ is required. The seasonally differenced series $\nabla_{12} X_t$ is the 12-month change in concentration. Its ACF still decays slowly (0.856, 0.792 and 0.745 at lags 1–3), and its level rises steadily over time as the growth rate accelerates, so it does not have a constant mean.

Augmented Dickey–Fuller tests. Because seasonal swings mimic mean reversion, the ADF test [4] was applied to the seasonally adjusted series (Table 4). With a single mean, τ ranges from 3.57 to 4.85, far above the 5% critical value of -2.86; with a trend, τ ranges from -0.88 to -0.62, against a critical value of -3.41. A unit root is never rejected. For $\nabla_{12} X_t$ the test rejects (τ from -7.91 to -5.37), but an AR(1)-plus-drift model for $\nabla_{12} X_t$ estimates $\hat{\phi} = 0.9993$ (SE 0.0010), practically a unit root. For the doubly differenced series $\nabla \nabla_{12} X_t$, the test rejects decisively at every lag (τ from -37.89 to -22.05). I therefore model $W_t = (1 - B)(1 - B^{12}) X_t$, which has 809 observations, as a stationary series (Figure 7).

Estimation versus differencing. As an alternative to differencing, I estimated the deterministic components by regression on a quadratic trend and 11 monthly dummy variables (Table 5). The fit is excellent ($R^2 = 0.9994$, residual SD 0.82 ppm), and the positive quadratic coefficient confirms that the trend is accelerating. The residuals, however, are far from white noise (Figure 8): their lag-1 autocorrelation is 0.915, $Q(12) = 5991$, and the Durbin–Watson statistic is $D = 0.17$, far below the value of 2 expected for independent errors. Deviations from the fixed polynomial persist for years, so differencing is the more appropriate treatment; a regression with ARMA errors is evaluated as a competing model in Section 4.

4. Model Selection and Model-Building Procedure

The model was built with the Box–Jenkins procedure [5] (plot, transform, identify, estimate, diagnose, forecast), with $d = D = 1$ and $T = 12$ fixed in Section 3.

White-noise test. The Ljung–Box test [6] for W_t gives $Q(6) = 69.99$, $Q(12) = 283.32$, $Q(18) = 295.43$ and $Q(24) = 301.19$ (all $p < 0.001$), so W_t is not white noise and must be modeled (Table 6).

Identification. The sample ACF of W_t (Figure 9) has a significant spike at lag 1 (-0.280) and a large spike at lag 12 (-0.471), with smaller satellite values at lags 11 (0.185) and 13 (0.042). Almost all other autocorrelations lie within the $\pm 1.96/\sqrt{809} = \pm 0.069$ bounds. The sample PACF, by contrast, tails off: at lags 1–3 it is -0.280, -0.146 and -0.135, and at the seasonal lags 12, 24 and 36 it is -0.378, -0.260 and -0.163. An ACF that cuts off together with a PACF that tails off is the signature of moving-average behavior [3]. Here that behavior appears at both the non-seasonal lag 1 and the seasonal lag 12. This points to the multiplicative seasonal ARIMA model [5]:

$$(1-B)(1-B^{12})X_t = (1-\theta_1 B)(1-\Theta_1 B^{12})Z_t, \qquad Z_t \sim \mathrm{WN}(0,\sigma_Z^2), \tag{1}$$

that is, ARIMA$(0,1,1)\times(0,1,1)_{12}$. Its right-hand side is an MA(13) with only two free parameters, which explains the small autocorrelations at lags 11 and 13.

Estimation and selection. Nine seasonal ARIMA candidates (Table 7) were estimated by maximum likelihood (ML) and compared with AIC and SBC [7]. ARIMA$(0,1,1)\times(0,1,1)_{12}$ attains the lowest SBC (465.39); ARIMA$(1,1,1)\times(0,1,1)_{12}$ has a lower AIC (447.58 versus 451.30) but a higher SBC (466.37). Pure AR alternatives fit worse, as expected when the PACF tails off rather than cutting off: ARIMA$(1,1,0)\times(0,1,1)_{12}$ and ARIMA$(2,1,0)\times(0,1,1)_{12}$ have SBC 486.92 and 479.42. Adding seasonal AR terms raises the SBC by at least 6.7, and ARIMA$(0,1,1)\times(1,1,0)_{12}$ is clearly inadequate (SBC 683.43; residuals fail the Ljung–Box test). Because SBC penalizes extra parameters more heavily in large samples, the two-parameter model was selected. Its implied autocorrelations, $\rho(1) = -0.338$, $\rho(12) = -0.495$ and $\rho(11) = \rho(13) = 0.167$, closely match the sample values above.

Comparison of the four model families. Four model families were fitted, each with its order chosen by SBC (Table 8 and Figure 10). (a) *ARMA*: the best non-seasonal model for W_t, ARMA(3,2), leaves a residual autocorrelation of -0.409 at lag 12 (Ljung–Box $p < 0.001$), because a non-seasonal ARMA cannot represent the seasonal MA structure. (b) *ARIMA*: the best ARIMA$(p,1,q)$ with drift, ARIMA(4,1,4), has four AR roots of modulus 1.000–1.001 with periods of 6 and 12 months. It imitates a seasonal difference with unit roots, and its residuals remain correlated at lag 12 (0.180, $p < 0.001$). (c) *Regression with autocorrelated errors*: the quadratic trend and monthly dummies with ARMA(2,1) errors give an AR root of 1.02, almost a unit root, and residual correlation of 0.138 at lag 12 ($Q(12)$ $p < 0.001$; Table 9). The standard remedy is then to difference both sides at lags 1 and 12, which removes the regressors and leads back to a seasonal ARIMA model. (d) *Seasonal ARIMA*: only ARIMA$(0,1,1)\times(0,1,1)_{12}$ leaves white-noise residuals (lag-12 autocorrelation -0.005, $Q(12)$ $p = 0.464$). The seasonal ARIMA family is therefore required, and the selected model is itself an ARMA model for W_t, namely an MA(13) restricted to two parameters.

5. Hypothesis Tests and Confidence Intervals

The ML estimates (Table 10) are $\hat\theta_1 = 0.3885$ (SE 0.0367) and $\hat\Theta_1 = 0.8700$ (SE 0.0170), with $\hat\sigma_Z^2 = 0.1012$, so $\hat\sigma_Z = 0.318$ ppm. The log-likelihood is -222.65. For $H_0 : \theta_1 = 0$, $z = 10.59$ ($p < 0.001$), with 95% confidence interval $[0.317, 0.460]$. For $H_0 : \Theta_1 = 0$, $z = 51.28$ ($p < 0.001$), with 95% interval $[0.837, 0.903]$. Both terms are therefore essential. Conditional least squares gives similar values, $\hat\theta_1 = 0.4172$ (SE 0.0360) and $\hat\Theta_1 = 0.8116$ (SE 0.0194). This shows that the conclusions do not depend on the estimation method.

The root of $1-\hat\theta_1 B$ is $1/\hat\theta_1 = 2.57$, and the roots of $1-\hat\Theta_1 B^{12}$ have modulus $\hat\Theta_1^{-1/12} = 1.012$. Both lie outside the unit circle, so W_t is *invertible*, and with no AR part it is also causal; the model

can therefore be used for forecasting. The interval for Θ_1 lies entirely below 1; $\Theta_1 = 1$ would indicate a deterministic seasonal pattern, so seasonal differencing is justified. The large $\hat{\Theta}_1$ means the seasonal pattern changes only slowly from year to year, and $\hat{\theta}_1$ means about 39% of each month's shock is reversed the following month.

6. Model Adequacy Checking and Remedial Methods

Independence. The residuals (Figure 11) pass the Ljung–Box test at all four lags examined (Table 11 and Figure 12): $Q(6) = 9.05$ ($p = 0.060$), $Q(12) = 9.74$ ($p = 0.464$), $Q(18) = 14.43$ ($p = 0.567$) and $Q(24) = 20.83$ ($p = 0.532$), with degrees of freedom reduced by the two estimated parameters. Only one of the first 36 residual autocorrelations lies outside ± 0.069: lag 3, at -0.081. About 1.8 are expected by chance.

Constant variance. The Ljung–Box test on the squared residuals gives $Q(12) = 12.40$ ($p = 0.414$) and $Q(24) = 35.36$ ($p = 0.063$). The residual variance is similar in both halves of the sample (0.091, 0.107), so it is constant.

Normality. The residuals are nearly symmetric (skewness 0.11, excess kurtosis 0.35), but three tests reject normality at 5%: Shapiro–Wilk $W = 0.9963$ ($p = 0.048$), Anderson–Darling ($p = 0.038$) and Jarque–Bera ($p = 0.006$). Five months have standardized residuals beyond ± 3 (Table 12): March 1959 (-3.46), March 1977 (3.31), September 1986 (3.11), April 2016 (3.97) and March 2022 (-3.53). The April 2016 value coincides with the unusually rapid CO_2 growth during the 2015–16 El Niño.

Remedial methods.

(i) *Lag-3 autocorrelation.* To remove this correlation, I added an AR(1) term. ARIMA$(1,1,1) \times (0,1,1)_{12}$ has a significant $\hat{\phi}_1 = 0.215$ (SE 0.084, $p = 0.010$) and reduces the lag-3 residual autocorrelation to -0.042. Its AR and MA roots (4.65 and 1.74) differ, so there is no parameter redundancy, but it raises the SBC and, in the hold-out of Section 7, forecast less accurately (RMSE 1.168 versus 1.110) with only 83% interval coverage against 97%. The parsimonious model was retained.

(ii) *Non-normality.* The five outlying months were treated as missing and the model was re-estimated by exact likelihood (Kalman filter). The estimates were almost unchanged (Table 13; $\hat{\theta}_1 = 0.3838$, $\hat{\Theta}_1 = 0.8689$), and normality was no longer rejected (Shapiro–Wilk $p = 0.211$, Anderson–Darling $p = 0.101$, Jarque–Bera $p = 0.094$). The residuals still passed the Ljung–Box tests ($p \geq 0.092$). The non-normality is thus caused by a few genuine but unusual months; with $n = 822$ the ML estimates are approximately normal, so the tests and intervals remain valid.

Because the residual variance is stable, no variance-stabilizing transformation is needed, and the fitted ARIMA$(0,1,1) \times (0,1,1)_{12}$ model is adequate.

7. Forecasting

Rolling-origin comparison. Each method was re-estimated at 23 forecast origins (every August from 2000 to 2022) and forecast the next 12 months, giving 276 out-of-sample forecasts per method (Figure 13 and Table 14). The seasonal ARIMA model achieved RMSE 0.552 ppm, with 95% interval coverage of 96.0%. It was essentially tied with Holt–Winters exponential smoothing [8] (0.543 ppm, 96.0%), and more accurate than regression with ARMA errors (0.580 ppm, 93.8%), the

ARMA model for W_t (0.677 ppm) and non-seasonal ARIMA (0.697 ppm). Holt–Winters estimates $\hat{\alpha} = 0.534$, $\hat{\beta} = 0.014$, $\hat{\gamma} = 0.299$ (Table 15).

Recent hold-out. Fitted to data through August 2023 and used to forecast September 2023 to August 2026 (Figure 14 and Table 16), the seasonal ARIMA model had RMSE 1.110 ppm, MAPE 0.24% and 97.2% interval coverage. Holt–Winters had RMSE 0.981 ppm, the trend-and-dummies regression 1.491 ppm (only 52.8% coverage, as it ignores the residual autocorrelation) and the seasonal naive method 6.448 ppm. All methods under-predicted 2024, when the annual mean rose by a record 3.52 ppm during the 2023–24 El Niño. The ARMA model for W_t scored best in this single window (0.655 ppm) only by carrying forward the latest annual increase.

Forecasts to 2030. The model estimated on all 822 months (Figure 15 and Table 17) gives a forecast for September 2026 of 426.16 ppm, with 95% prediction interval [425.53, 426.78]. For the May 2027 seasonal peak it gives 434.36 ppm [433.11, 435.60]. The forecast first exceeds 440 ppm in April 2030 (440.84 ppm [437.78, 443.89]) and peaks at 441.59 ppm [438.49, 444.69] in May 2030. The forecast for December 2030 is 439.03 ppm [435.59, 442.47]. Forecast annual means rise from 431.44 ppm (2027) to 438.67 ppm (2030), about 2.4 ppm per year, and the 95% interval widens from 1.25 ppm to 6.88 ppm, reflecting the accumulated uncertainty of a doubly integrated process.

8. Conclusions

Monthly CO_2 at Mauna Loa is a non-stationary series with an accelerating stochastic trend and a stable additive seasonal cycle. Of the four model families compared (ARMA, ARIMA, seasonal ARIMA, and regression with autocorrelated errors), only the seasonal ARIMA leaves white-noise residuals and gives the most accurate rolling forecasts, and the ACF/PACF and information criteria point to the parsimonious ARIMA$(0,1,1) \times (0,1,1)_{12}$. CO_2 is not only increasing but increasing faster, from under 1 ppm per year in the 1960s to about 2.5 ppm per year today. The model projects that concentrations will cross 440 ppm in spring 2030. As a univariate model it cannot anticipate El Niño-driven surges such as those of 2016 and 2024; adding climate indices or emissions data is a natural extension.

Supplementary material. Figures and tables are in the Appendix; R 4.3.3 code is provided separately.

References

[1] Charles D. Keeling. The concentration and isotopic abundances of carbon dioxide in the atmosphere. *Tellus*, 12(2): 200–203, 1960. doi:10.1111/j.2153-3490.1960.tb01300.x.

[2] Xin Lan and Ralph F. Keeling. Trends in atmospheric carbon dioxide: Mauna Loa monthly mean data. NOAA Global Monitoring Laboratory, 2026. URL https://gml.noaa.gov/ccgg/trends/data.html.

[3] Peter J. Brockwell and Richard A. Davis. *Introduction to Time Series and Forecasting*. Springer, Cham, 3rd edition, 2016. doi:10.1007/978-3-319-29854-2.

[4] David A. Dickey and Wayne A. Fuller. Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association*, 74(366):427–431, 1979. doi:10.1080/01621459.1979.10482531.

[5] George E. P. Box, Gwilym M. Jenkins, Gregory C. Reinsel, and Greta M. Ljung. *Time Series Analysis: Forecasting and Control*. Wiley, Hoboken, NJ, 5th edition, 2015. ISBN 978-1-118-67502-1. URL https://search.worldcat.org/title/908107438.

[6] Greta M. Ljung and George E. P. Box. On a measure of lack of fit in time series models. *Biometrika*, 65(2):297–303, 1978. doi:10.1093/biomet/65.2.297.

[7] Gideon Schwarz. Estimating the dimension of a model. *The Annals of Statistics*, 6(2):461–464, 1978. doi:10.1214/aos/1176344136.

[8] Peter R. Winters. Forecasting sales by exponentially weighted moving averages. *Management Science*, 6(3):324–342, 1960. doi:10.1287/mnsc.6.3.324.

A. Figures

The figures are numbered in the order in which the text refers to them. Click a figure or table number in the text to jump here.

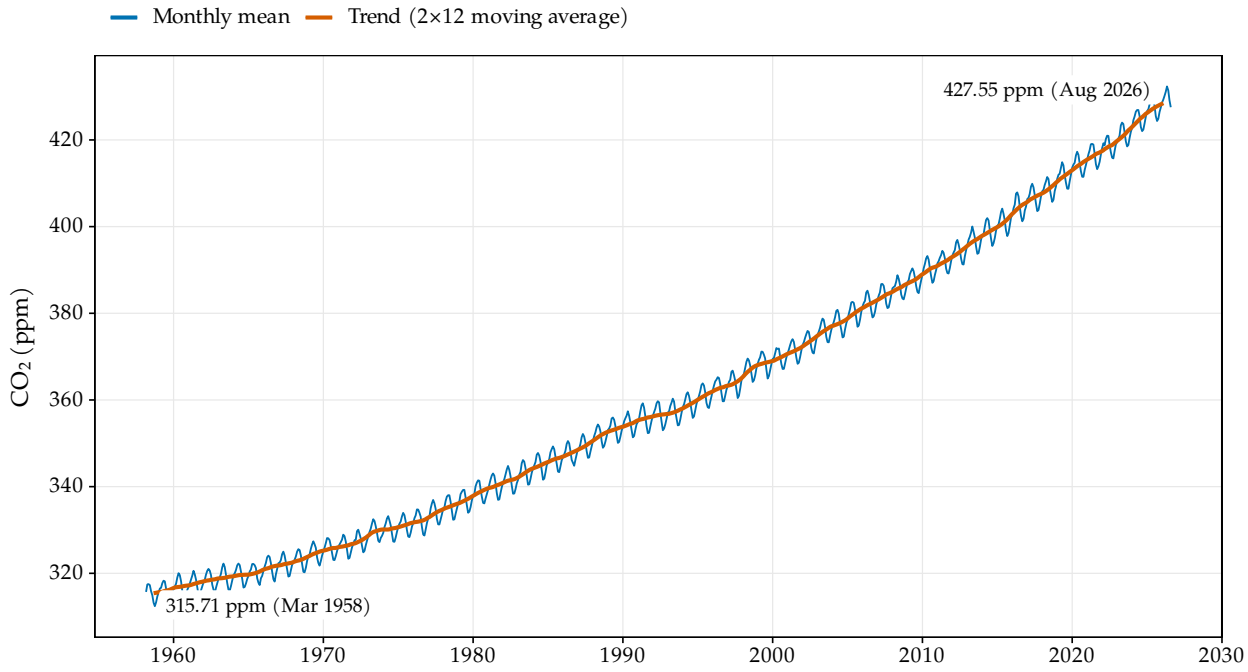


Figure 1. Monthly mean atmospheric CO_2 at Mauna Loa, March 1958 to August 2026, with the 2×12 moving-average trend.

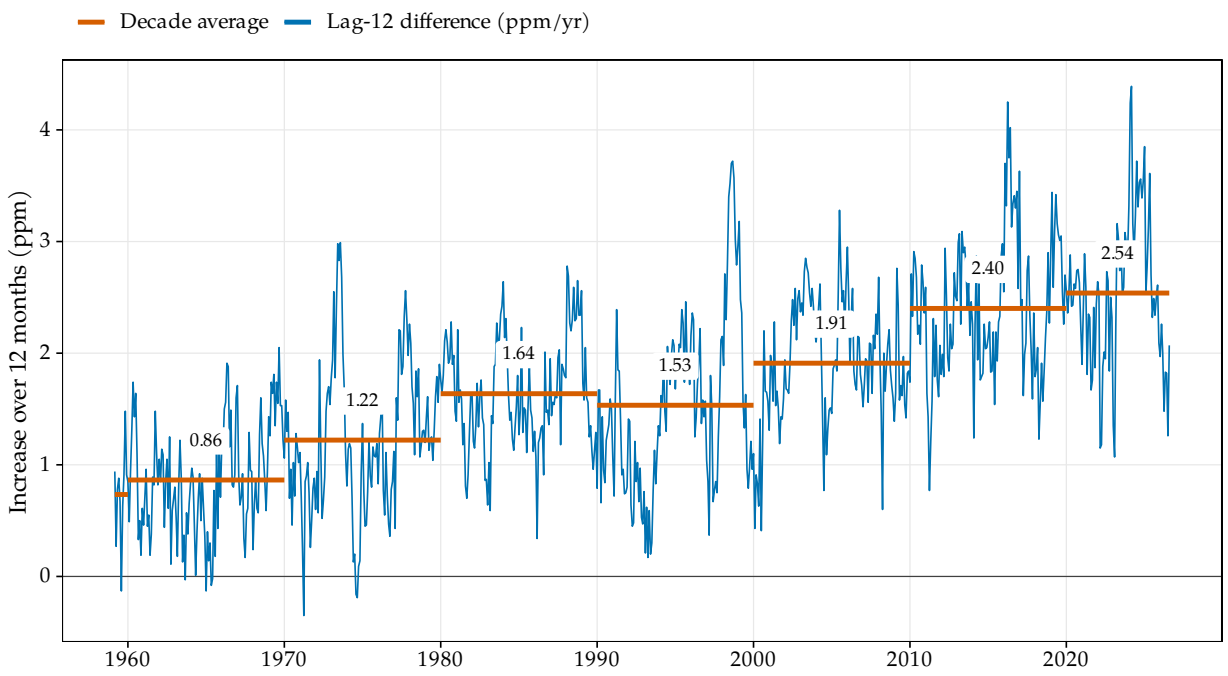


Figure 2. Twelve-month change $\nabla_{12} X_t$ (ppm per year) with decade averages: growth tripled from the 1960s to the 2020s.

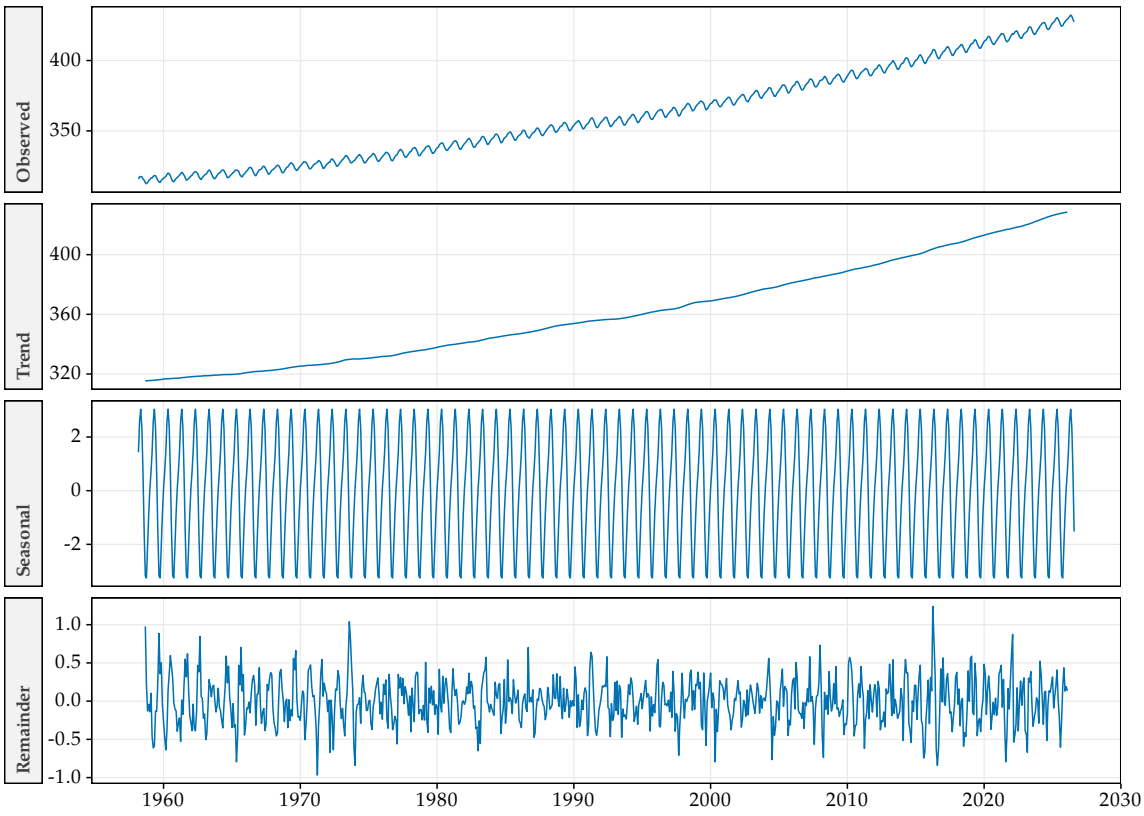


Figure 3. Classical additive decomposition of X_t into trend, seasonal and remainder components.

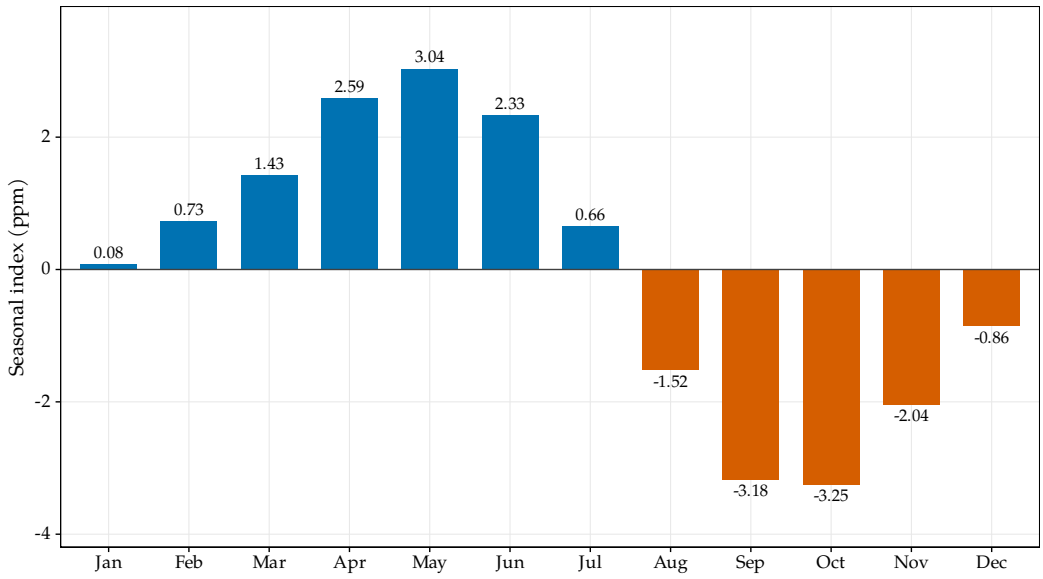


Figure 4. Monthly seasonal indices from the additive decomposition: peak in May (+3.04 ppm), trough in October (−3.25 ppm).

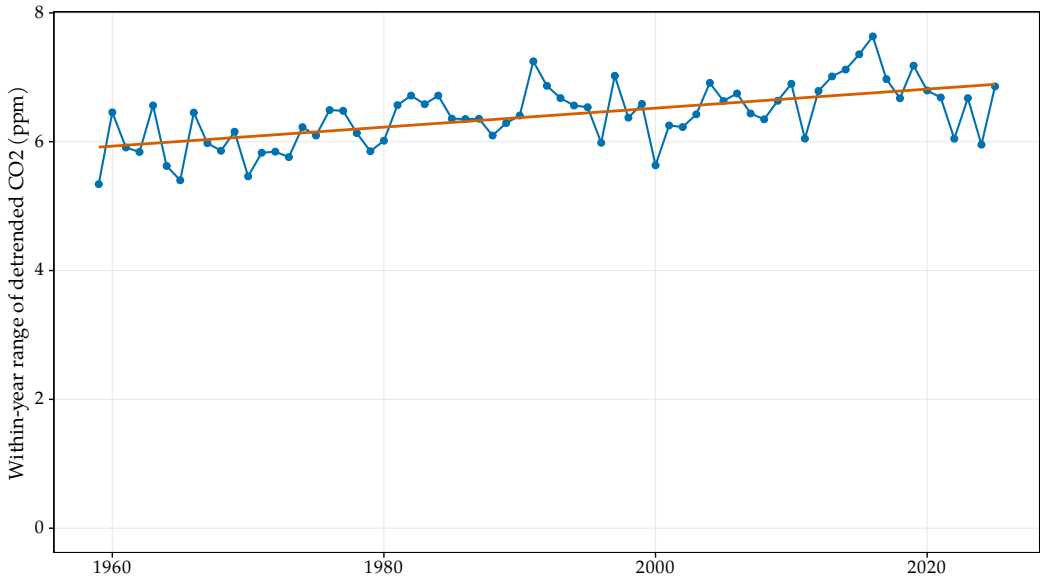


Figure 5. Within-year range of the detrended series by year, with a linear trend. The amplitude does not grow in proportion to the level, so no transformation is needed.

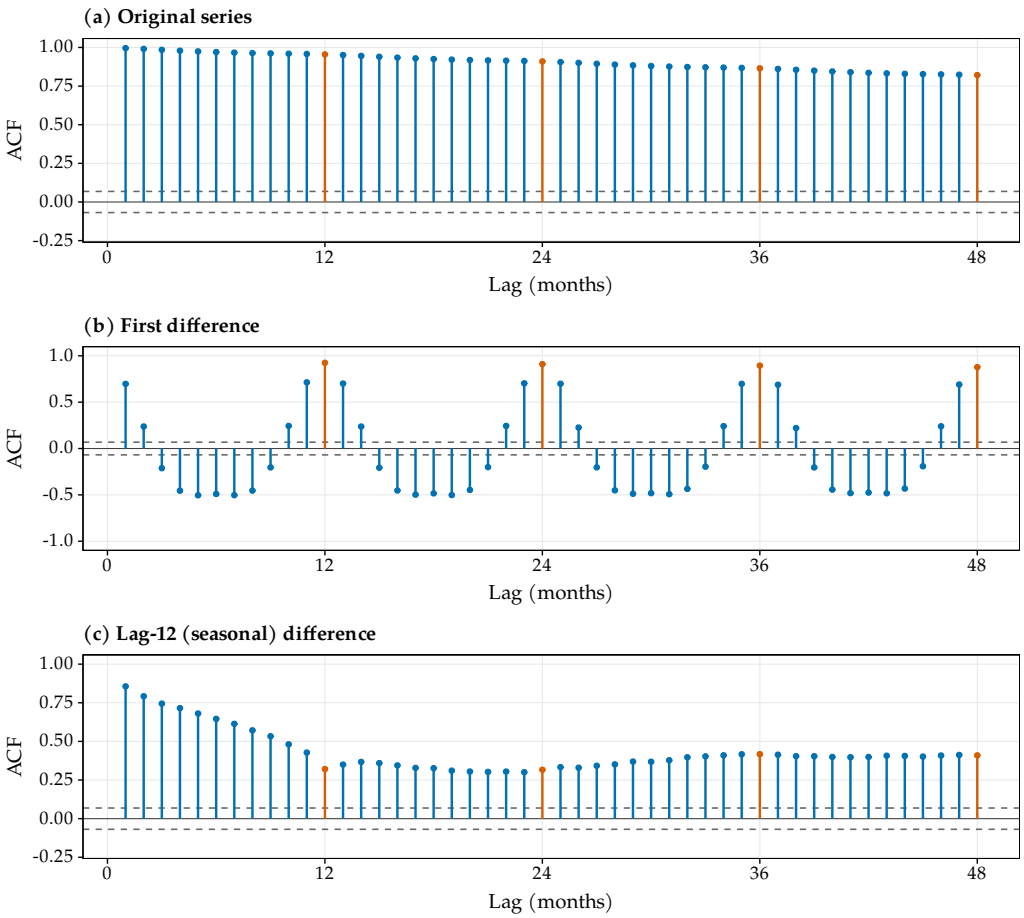


Figure 6. Sample ACF of (a) X_t, (b) ∇X_t and (c) $\nabla_{12} X_t$. Seasonal lags are highlighted; dashed lines are the $\pm 1.96/\sqrt{n}$ bounds.

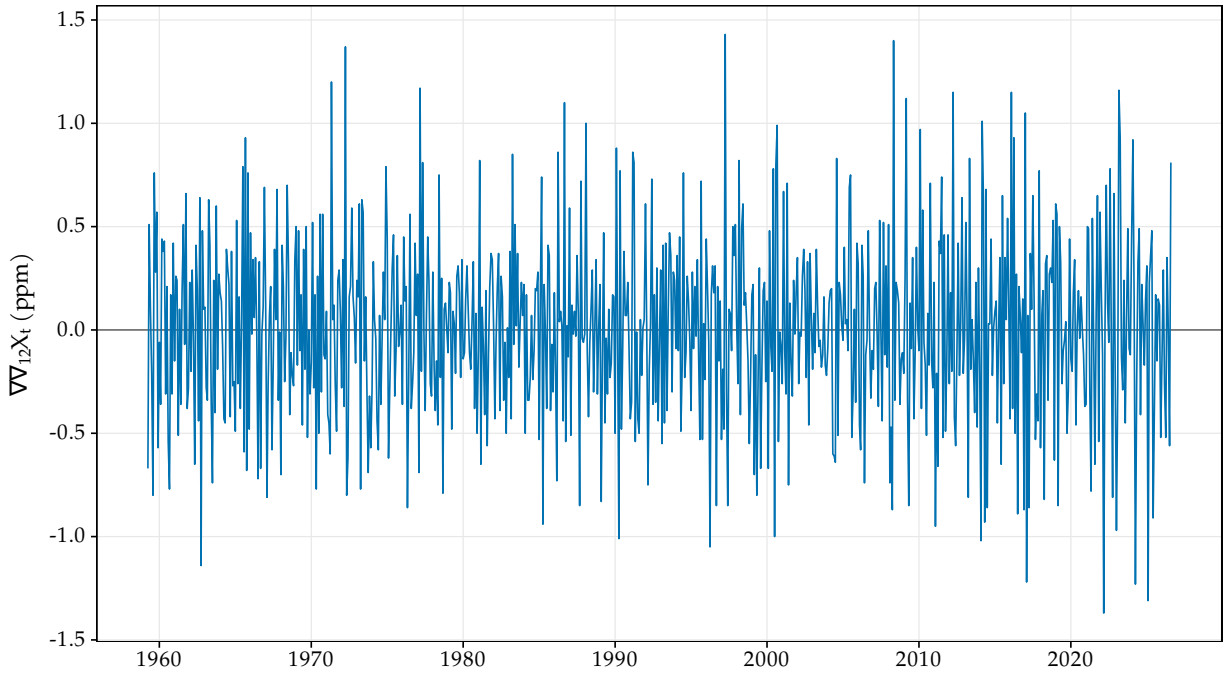


Figure 7. The stationary series $W_t = \nabla\nabla_{12}X_t$ used for model identification.

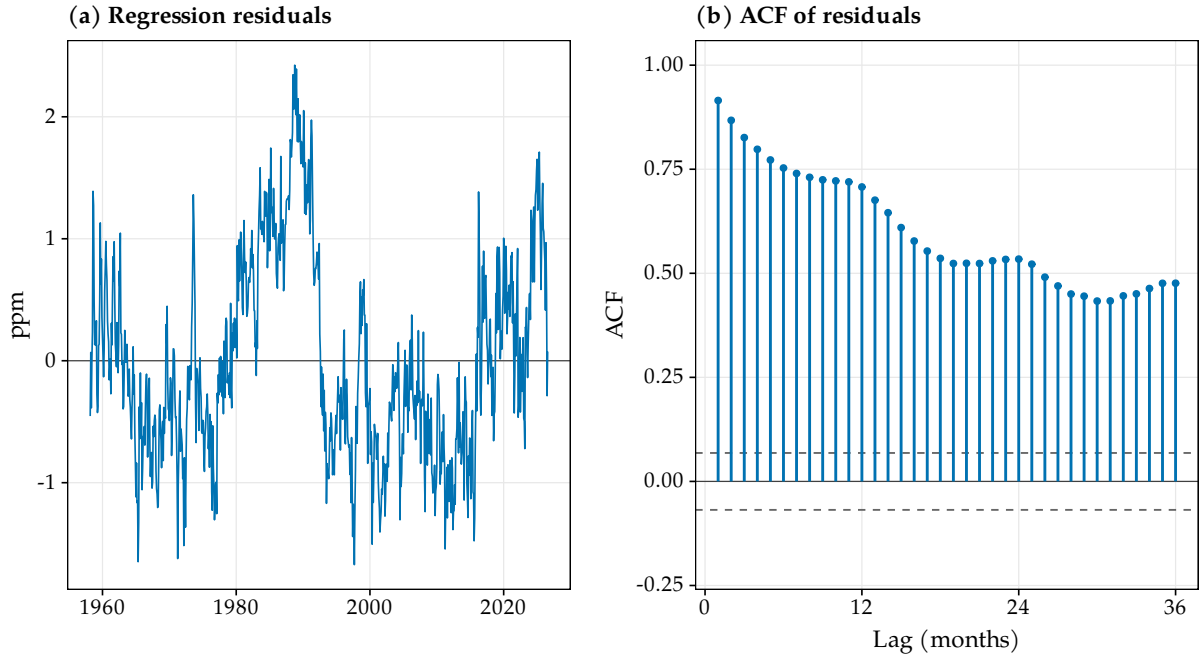


Figure 8. Regression on a quadratic trend and monthly dummies: (a) residuals and (b) their sample ACF. The residuals persist for years.

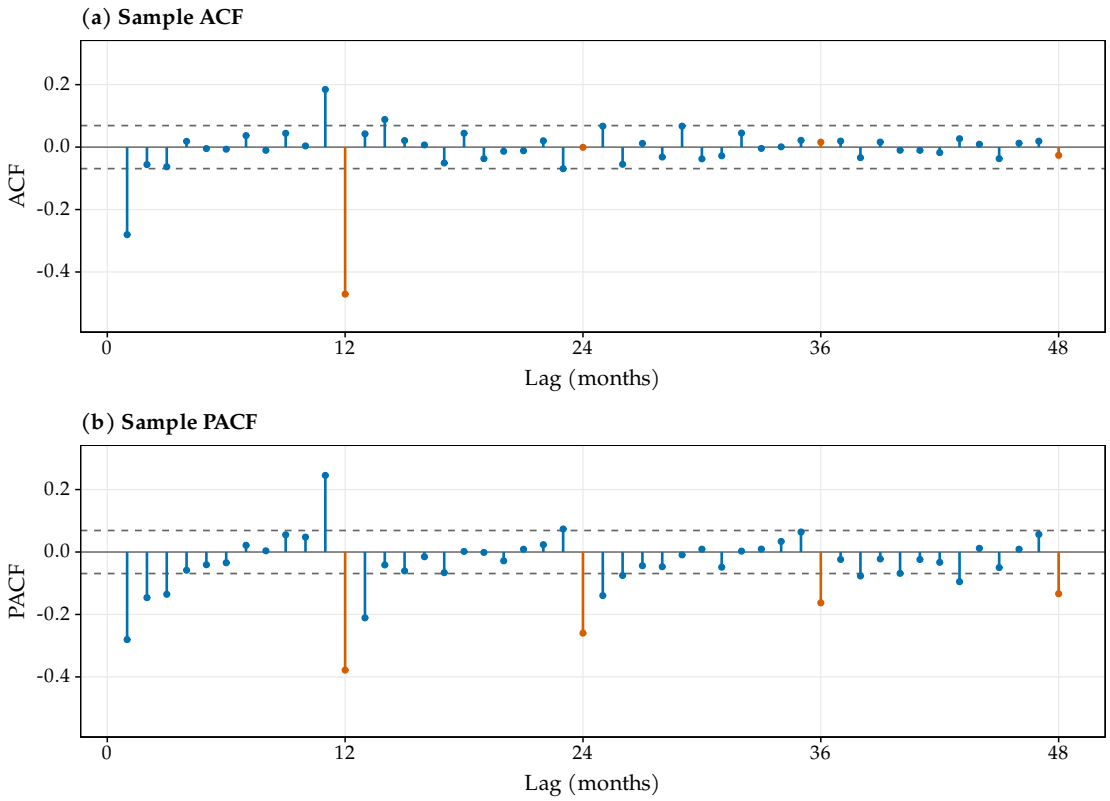


Figure 9. Sample ACF (a) and PACF (b) of W_t. The ACF cuts off after lags 1 and 12; the PACF tails off.

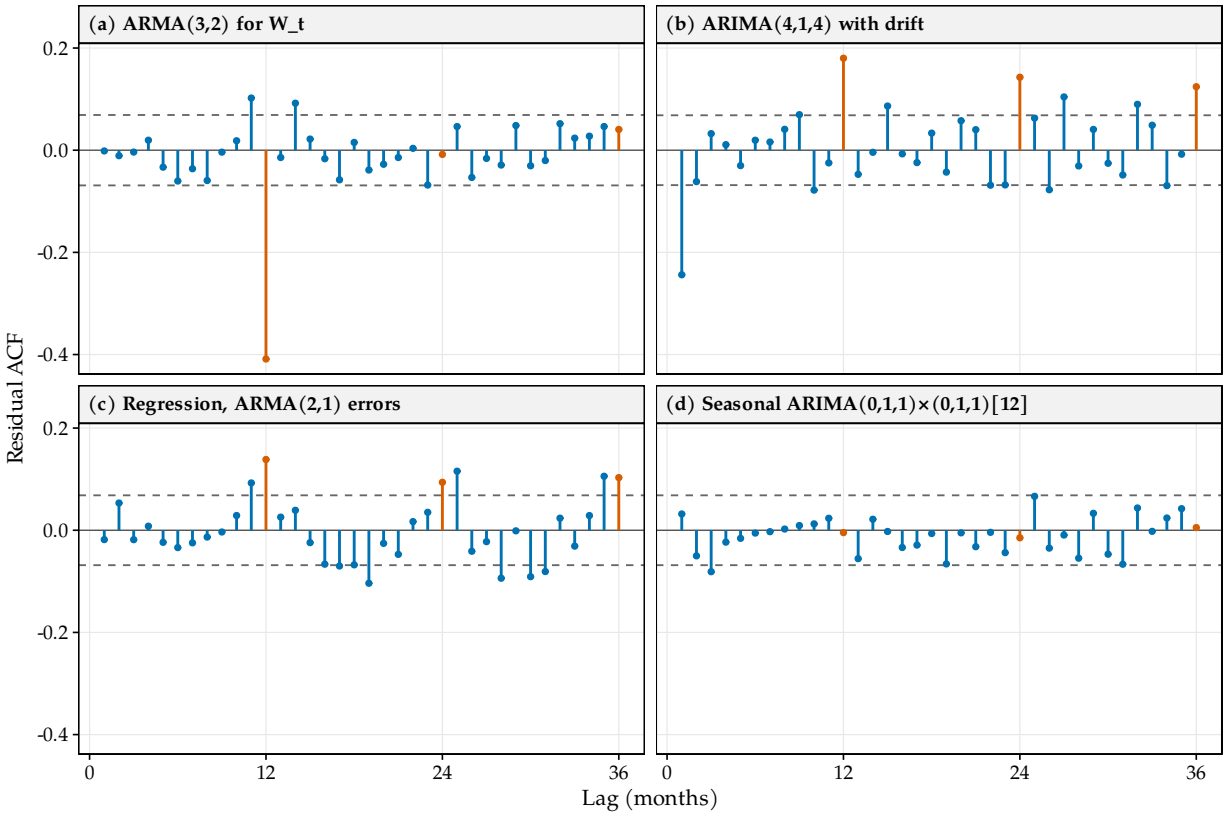


Figure 10. Residual ACF of the best model in each of the four families. Only the seasonal ARIMA (d) has no spikes at the seasonal lags.

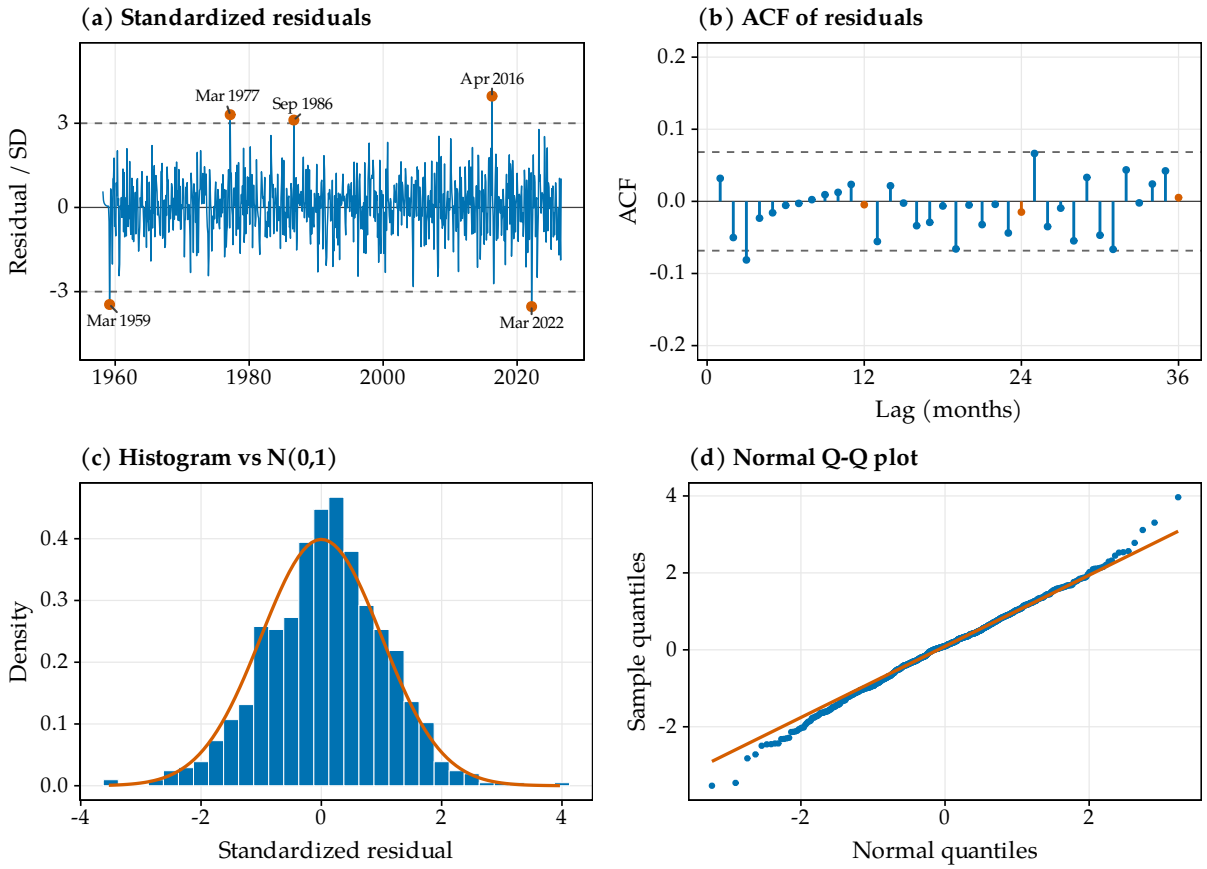


Figure 11. Residual diagnostics for ARIMA$(0,1,1) \times (0,1,1)_{12}$: (a) standardized residuals with the five months beyond ± 3 labeled, (b) residual ACF, (c) histogram with the $N(0,1)$ density, (d) normal Q–Q plot.

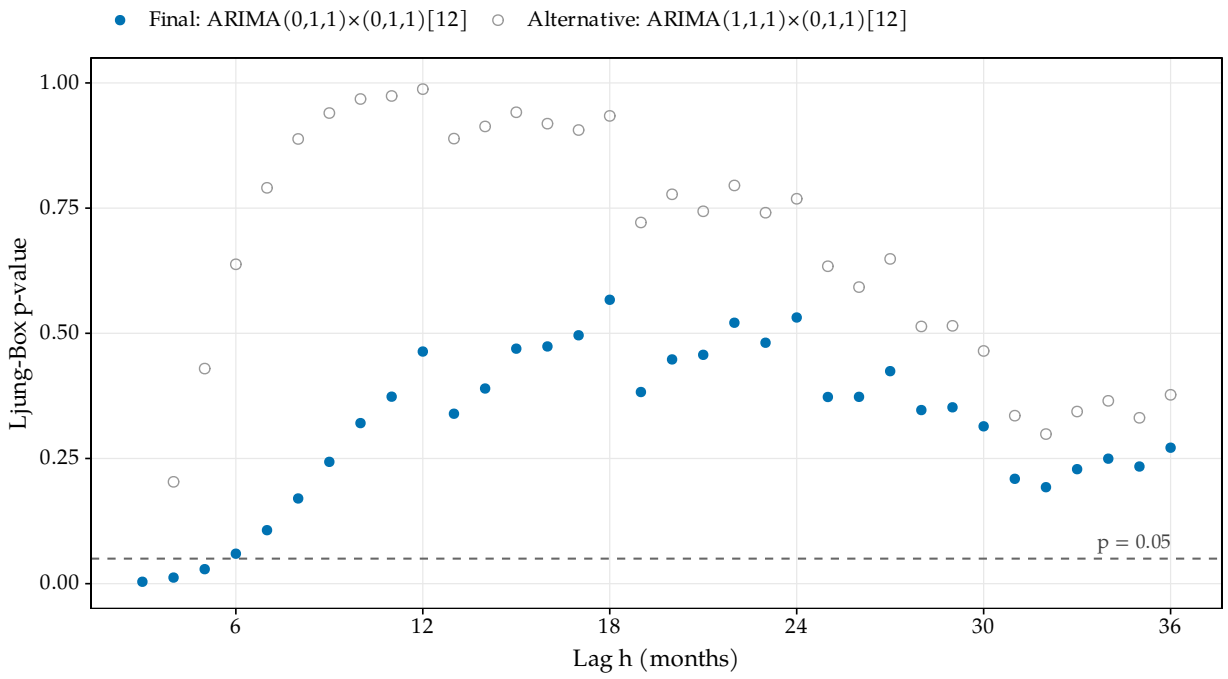


Figure 12. Ljung–Box p-values by lag for the final and the alternative model; the dashed line is $p = 0.05$.

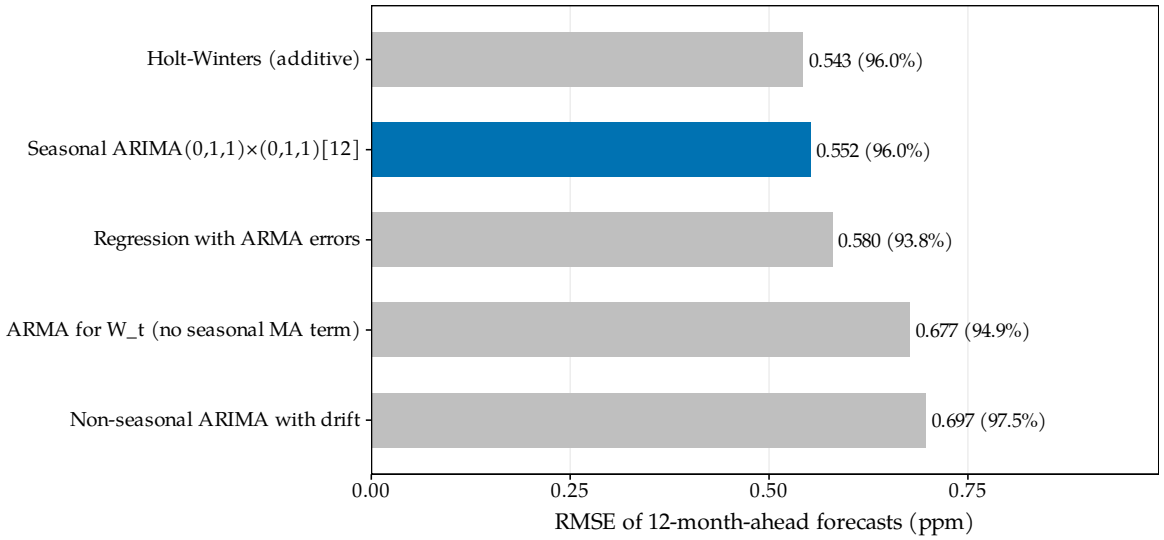


Figure 13. Rolling-origin evaluation: RMSE of 12-month-ahead forecasts from 23 origins (276 forecasts per method), with 95% interval coverage in parentheses.

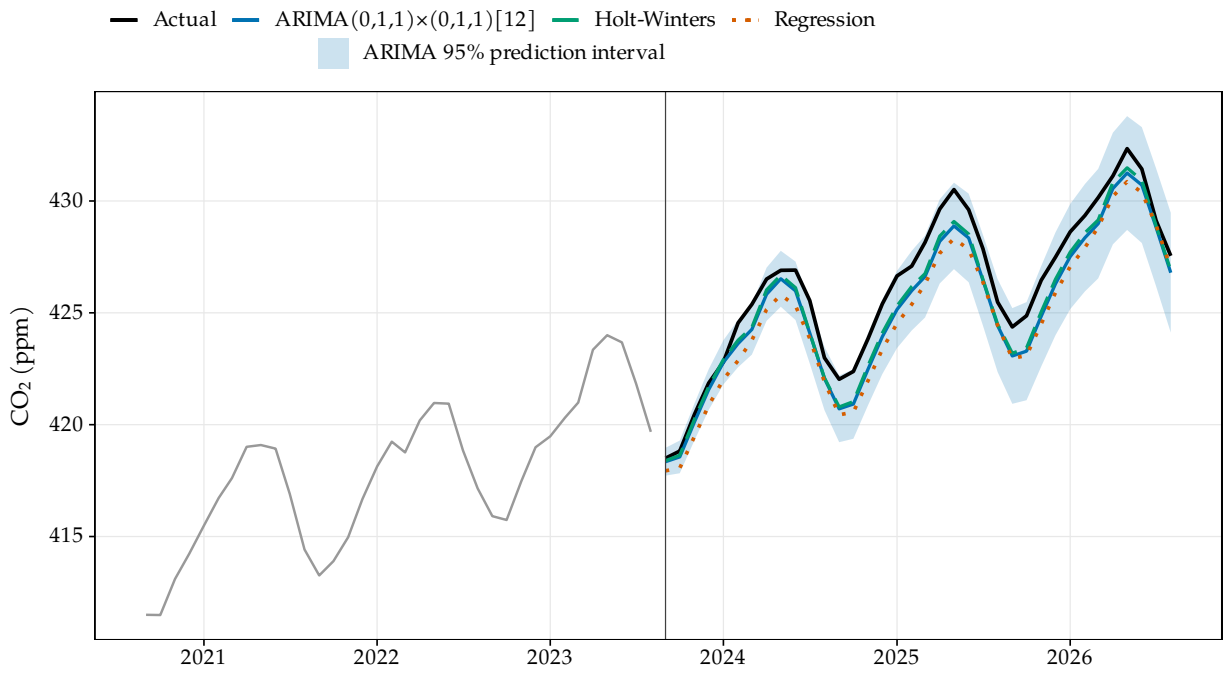


Figure 14. Hold-out forecasts for September 2023 to August 2026 from models fitted through August 2023, with the ARIMA 95% prediction interval.

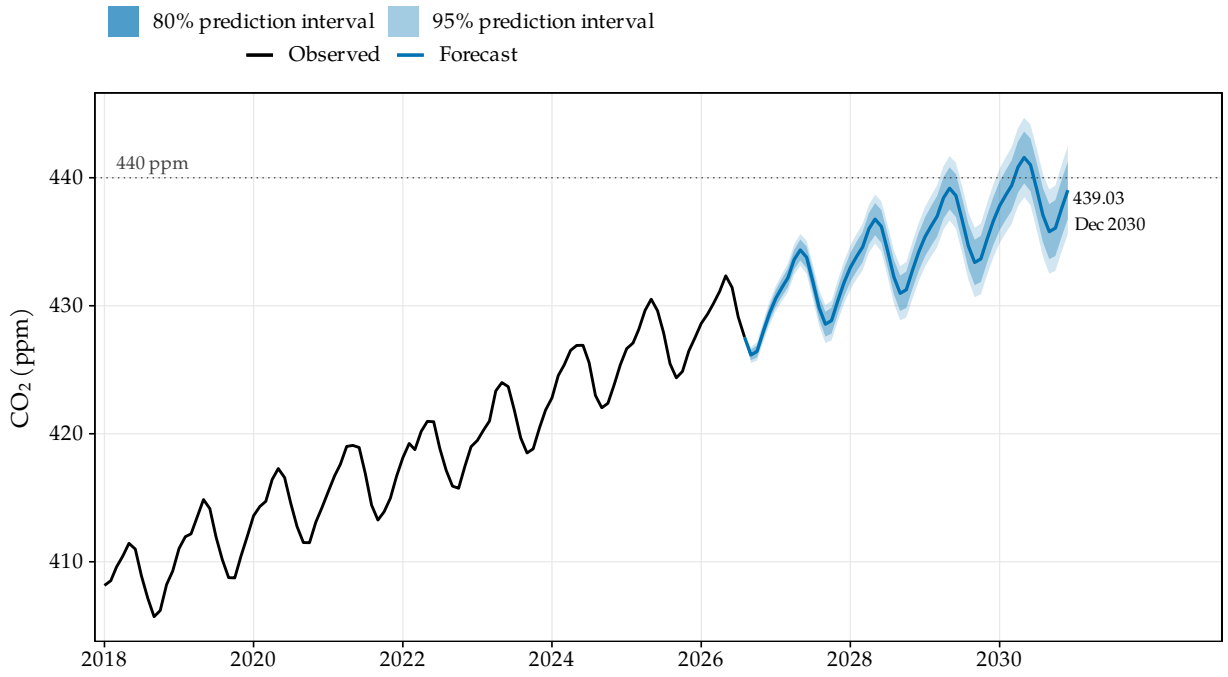


Figure 15. Forecast of monthly CO_2 to December 2030 from ARIMA$(0,1,1) \times (0,1,1)_{12}$, with 80% and 95% prediction intervals.

B. Tables

Table 1. Descriptive statistics of the monthly CO_2 series.

Statistic	Value
Observations (months)	822
First month	Mar 1958
Last month	Aug 2026
Mean (ppm)	361.36
Standard deviation (ppm)	33.46
Minimum (ppm)	312.42 (Oct 1958)
Maximum (ppm)	432.34 (May 2026)
Average annual increase (ppm/yr)	1.68

Table 2. Mean level, mean annual increase and seasonal amplitude by decade.

Decade	Mean CO_2 (ppm)	Annual increase (ppm/yr)	Seasonal amplitude (ppm)
1950s	315.64	0.73	5.34
1960s	320.29	0.86	6.02
1970s	330.86	1.22	6.02
1980s	345.65	1.64	6.41
1990s	360.58	1.53	6.63
2000s	378.77	1.91	6.43
2010s	400.41	2.40	6.97
2020s	421.32	2.54	6.50

Table 3. Monthly seasonal indices from the additive decomposition (ppm).

Month	Seasonal index (ppm)
Jan	0.083
Feb	0.729
Mar	1.433
Apr	2.587
May	3.036
Jun	2.334
Jul	0.656
Aug	−1.525
Sep	−3.177
Oct	−3.253
Nov	−2.043
Dec	−0.861

Table 4. Augmented Dickey–Fuller tests (H_0: unit root). Seasonally adjusted series at lags 0–2; reject H_0 when τ is below the critical value.

Series	Type	Lags	τ	5% critical value	1% critical value	Decision
Seasonally adjusted X_t	Single mean	0	3.57	−2.86	−3.43	Fail to reject (unit root)
Seasonally adjusted X_t	Single mean	1	4.53	−2.86	−3.43	Fail to reject (unit root)
Seasonally adjusted X_t	Single mean	2	4.85	−2.86	−3.43	Fail to reject (unit root)
Seasonally adjusted X_t	Trend	0	−0.88	−3.41	−3.96	Fail to reject (unit root)
Seasonally adjusted X_t	Trend	1	−0.79	−3.41	−3.96	Fail to reject (unit root)
Seasonally adjusted X_t	Trend	2	−0.62	−3.41	−3.96	Fail to reject (unit root)
X_t (13 augmentation lags)	Trend	13	−0.33	−3.41	−3.96	Fail to reject (unit root)
Lag-12 difference	Single mean	0	−7.91	−2.86	−3.43	Reject H_0 (stationary)
Lag-12 difference	Single mean	1	−6.14	−2.86	−3.43	Reject H_0 (stationary)
Lag-12 difference	Single mean	2	−5.37	−2.86	−3.43	Reject H_0 (stationary)
Regular + seasonal difference	Single mean	0	−37.89	−2.86	−3.43	Reject H_0 (stationary)
Regular + seasonal difference	Single mean	1	−26.26	−2.86	−3.43	Reject H_0 (stationary)
Regular + seasonal difference	Single mean	2	−22.05	−2.86	−3.43	Reject H_0 (stationary)

Table 5. Regression of X_t on a quadratic trend and monthly dummies (estimation approach), with R^2 and the Durbin–Watson statistic.

Term	Estimate	SE	t	p
Intercept	314.7597	0.1283	2,453.05	< 0.001
t	0.0621	0.0005	128.89	< 0.001
t^2	0.0001	0.0000	165.35	< 0.001
Month: Feb	0.6444	0.1403	4.59	< 0.001
Month: Mar	1.3417	0.1398	9.60	< 0.001
Month: Apr	2.4979	0.1398	17.87	< 0.001
Month: May	2.9496	0.1398	21.10	< 0.001
Month: Jun	2.2487	0.1398	16.09	< 0.001
Month: Jul	0.5630	0.1398	4.03	< 0.001
Month: Aug	−1.5948	0.1398	−11.41	< 0.001
Month: Sep	−3.2629	0.1403	−23.26	< 0.001
Month: Oct	−3.3378	0.1403	−23.80	< 0.001
Month: Nov	−2.1275	0.1403	−15.17	< 0.001
Month: Dec	−0.9436	0.1403	−6.73	< 0.001
R^2	0.9994			
Durbin-Watson D	0.1693			

Table 6. Ljung–Box tests for $W_t = \nabla\nabla_{12}X_t$ (H_0: white noise).

Lag	Q	p
6	69.99	< 0.001
12	283.32	< 0.001
18	295.43	< 0.001
24	301.19	< 0.001

Table 7. Candidate seasonal ARIMA models estimated by maximum likelihood, sorted by SBC.

Model	Parameters	Log-likelihood	AIC	SBC	ΔSBC	Ljung–Box $Q(24)$ p
$\text{ARIMA}(0,1,1)\times(0,1,1)_{12}$	2	−222.65	451.30	465.39	0.00	0.532
$\text{ARIMA}(1,1,1)\times(0,1,1)_{12}$	3	−219.79	447.58	466.37	0.98	0.769
$\text{ARIMA}(0,1,2)\times(0,1,1)_{12}$	3	−220.30	448.59	467.38	1.99	0.723
$\text{ARIMA}(0,1,1)\times(0,1,2)_{12}$	3	−222.65	453.30	472.08	6.69	0.472
$\text{ARIMA}(0,1,1)\times(1,1,1)_{12}$	3	−222.65	453.30	472.08	6.69	0.472
$\text{ARIMA}(1,1,1)\times(1,1,1)_{12}$	4	−219.79	449.57	473.05	7.66	0.715
$\text{ARIMA}(2,1,0)\times(0,1,1)_{12}$	3	−226.32	460.64	479.42	14.03	0.213
$\text{ARIMA}(1,1,0)\times(0,1,1)_{12}$	2	−233.41	472.83	486.92	21.53	0.038
$\text{ARIMA}(0,1,1)\times(1,1,0)_{12}$	2	−331.67	669.34	683.43	218.04	<0.001

Table 8. Best model in each of the four families (k = number of ARMA parameters): residual checks and 36-month hold-out accuracy.

				Ljung–Box p		Hold-out	
Family	Best model	k	ACF lag 12	$Q(12)$	$Q(24)$	RMSE	Coverage (%)
ARMA	ARMA(3,2) for W_t	5	−0.409	<0.001	<0.001	0.655	100.0
ARIMA	ARIMA(4,1,4) with drift	8	0.180	<0.001	<0.001	2.127	100.0
Regression with ARMA errors	Trend + dummies, ARMA(2,1) errors	3	0.138	<0.001	<0.001	1.207	55.6
Seasonal ARIMA	$\text{ARIMA}(0,1,1)\times(0,1,1)_{12}$	2	−0.005	0.464	0.532	1.110	97.2

Table 9. Regression with ARMA errors: the AR coefficients sum to nearly 1 and the residuals fail the Ljung–Box test.

Error model	Sum of AR coefficients	Ljung–Box $Q(12)$ p	Ljung–Box $Q(24)$ p
ARMA(1,0)	0.914	<0.001	<0.001
ARMA(2,0)	0.930	<0.001	<0.001
ARMA(2,1)	0.990	<0.001	<0.001

Table 10. Parameter estimates, z-tests and 95% confidence intervals (maximum likelihood and conditional least squares).

Model	Method	Parameter	Estimate	SE	z	p	95% CI
Final: $\text{ARIMA}(0,1,1)\times(0,1,1)_{12}$	ML	θ_1 (MA(1))	0.3885	0.0367	10.59	<0.001	[0.317, 0.460]
Final: $\text{ARIMA}(0,1,1)\times(0,1,1)_{12}$	ML	Θ_1 (seasonal MA(1))	0.8700	0.0170	51.28	<0.001	[0.837, 0.903]
Final: $\text{ARIMA}(0,1,1)\times(0,1,1)_{12}$	CLS	θ_1 (MA(1))	0.4172	0.0360	11.60	<0.001	[0.347, 0.488]
Final: $\text{ARIMA}(0,1,1)\times(0,1,1)_{12}$	CLS	Θ_1 (seasonal MA(1))	0.8116	0.0194	41.84	<0.001	[0.774, 0.850]
Alternative: $\text{ARIMA}(1,1,1)\times(0,1,1)_{12}$	ML	ϕ_1 (AR(1))	0.2151	0.0837	2.57	0.010	[0.051, 0.379]
Alternative: $\text{ARIMA}(1,1,1)\times(0,1,1)_{12}$	ML	θ_1 (MA(1))	0.5744	0.0701	8.20	<0.001	[0.437, 0.712]
Alternative: $\text{ARIMA}(1,1,1)\times(0,1,1)_{12}$	ML	Θ_1 (seasonal MA(1))	0.8699	0.0169	51.50	<0.001	[0.837, 0.903]

Table 11. Residual diagnostics for the final model, the final model with the five outlying months set to missing, and the alternative ARIMA$(1,1,1)\times(0,1,1)_{12}$.

	Final model		Outlying months missing		Alternative model	
Test	Statistic	p	Statistic	p	Statistic	p
Ljung–Box $Q(6)$	9.050	0.060	7.985	0.092	1.696	0.638
Ljung–Box $Q(12)$	9.740	0.464	10.306	0.414	2.213	0.988
Ljung–Box $Q(18)$	14.427	0.567	14.871	0.534	7.730	0.934
Ljung–Box $Q(24)$	20.825	0.532	18.732	0.662	16.017	0.769
Ljung–Box $Q(12)$, squared	12.398	0.414	16.935	0.152	9.987	0.617
Ljung–Box $Q(24)$, squared	35.359	0.063	40.293	0.020	32.456	0.116
Shapiro–Wilk	0.996	0.048	0.997	0.211	0.996	0.037
Anderson–Darling	0.800	0.038	0.630	0.101	0.771	0.045
Jarque–Bera	10.306	0.006	4.735	0.094	11.132	0.004

Table 12. Months with standardized residuals beyond ± 3.

Month	Observed (ppm)	Standardized residual
Mar 1959	316.65	-3.46
Mar 1977	334.96	3.31
Sep 1986	345.48	3.11
Apr 2016	407.60	3.97
Mar 2022	418.76	-3.53

Table 13. Sensitivity of the estimates to the five outlying months.

Data	θ_1	Θ_1	$\hat{\sigma}_Z^2$
All months	0.3885	0.8700	0.1012
Outlying months set to missing	0.3838	0.8689	0.0978

Table 14. Rolling-origin accuracy of 12-month-ahead forecasts (23 origins, August 2000–2022).

Method	RMSE	MAE	95% PI coverage (%)	Forecasts
Holt–Winters (additive)	0.543	0.403	96.0	276
Seasonal ARIMA$(0,1,1)\times(0,1,1)_{12}$	0.552	0.411	96.0	276
Regression with ARMA errors	0.580	0.445	93.8	276
ARMA for W_t (no seasonal MA term)	0.677	0.521	94.9	276
Non-seasonal ARIMA with drift	0.697	0.543	97.5	276

Table 15. Holt–Winters smoothing parameters, full sample.

Parameter	Value
α (level)	0.5337
β (trend)	0.0139
γ (seasonal)	0.2990
Final level (ppm)	429.1464
Final slope (ppm/month)	0.2030

Table 16. Hold-out forecast accuracy, September 2023 to August 2026.

Method	RMSE	MAE	MAPE (%)	95% PI coverage (%)
Final: ARIMA$(0,1,1)\times(0,1,1)_{12}$	1.110	1.010	0.237	97.2
ARMA for W_t (no seasonal MA term)	0.655	0.501	0.117	100.0
Non-seasonal ARIMA with drift	2.127	1.930	0.452	100.0
Regression with ARMA errors	1.207	1.113	0.261	55.6
Alternative: ARIMA$(1,1,1)\times(0,1,1)_{12}$	1.168	1.072	0.251	83.3
Holt–Winters (additive)	0.981	0.875	0.205	97.2
Regression: quadratic trend + seasonal dummies	1.491	1.413	0.331	52.8
Seasonal naive	6.448	6.069	1.422	25.0

Table 17. Forecasts of monthly CO_2 (ppm) with 80% and 95% prediction intervals and the Holt–Winters forecast.

Month	Forecast	80% PI	95% PI	Holt–Winters
Sep 2026	426.16	[425.75, 426.56]	[425.53, 426.78]	426.19
Dec 2026	429.39	[428.80, 429.99]	[428.49, 430.30]	429.41
May 2027	434.36	[433.54, 435.17]	[433.11, 435.60]	434.47
Dec 2027	431.80	[430.72, 432.89]	[430.14, 433.46]	431.85
May 2028	436.77	[435.51, 438.02]	[434.85, 438.69]	436.90
Dec 2028	434.21	[432.72, 435.70]	[431.93, 436.49]	434.28
May 2029	439.18	[437.53, 440.82]	[436.66, 441.70]	439.34
Dec 2029	436.62	[434.75, 438.49]	[433.76, 439.49]	436.72
May 2030	441.59	[439.56, 443.61]	[438.49, 444.69]	441.78
Dec 2030	439.03	[436.78, 441.28]	[435.59, 442.47]	439.16